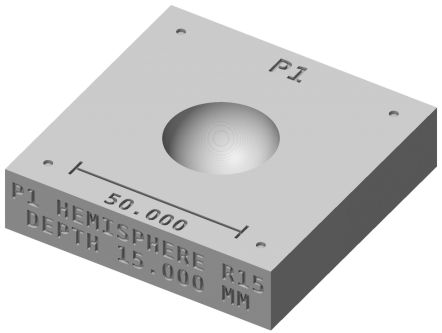
Wound phantoms: match the part to its picture

Drawn from the STEP files at one scale, cavity up, key line nearest. Sizes in mm and cm (1 cm³ = 1 mL); the mm depth, area and volume are as engraved on the parts. Pages 3 and 4 have the full detail.

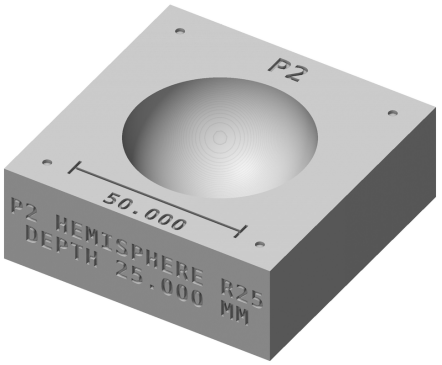


P1 Hemisphere

P1_hemisphere_r15.step

Opening Ø30.000 mm
Opening Ø3.0000 cm
Block 80 × 80 × 19 mm · 8.0 × 8.0 × 1.9 cm

Depth	15.000 mm	1.5000 cm
Area	706.86 mm²	7.0686 cm²
Volume	7068.58 mm³	7.06858 cm³
Perimeter	94.25 mm	9.425 cm



P2 Hemisphere

P2_hemisphere_r25.step

Opening Ø50.000 mm
Opening Ø5.0000 cm
Block 80 × 80 × 29 mm · 8.0 × 8.0 × 2.9 cm

Depth	25.000 mm	2.5000 cm
Area	1963.50 mm²	19.6350 cm²
Volume	32724.92 mm³	32.72492 cm³
Perimeter	157.08 mm	15.708 cm

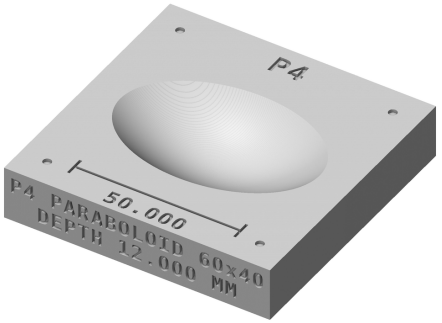


P3 Cone

P3_cone_r20_d10.step

Opening Ø40.000 mm
Opening Ø4.0000 cm
Block 80 × 80 × 14 mm · 8.0 × 8.0 × 1.4 cm

Depth	10.000 mm	1.0000 cm
Area	1256.64 mm²	12.5664 cm²
Volume	4188.79 mm³	4.18879 cm³
Perimeter	125.66 mm	12.566 cm



P4 Paraboloid

P4_paraboloid_30x20_d12.step

Opening 60.000 × 40.000 mm oval
Opening 6.0000 × 4.0000 cm
Block 80 × 80 × 16 mm · 8.0 × 8.0 × 1.6 cm

Depth	12.000 mm	1.2000 cm
Area	1884.96 mm²	18.8496 cm²
Volume	11309.73 mm³	11.30973 cm³
Perimeter	158.65 mm	15.865 cm

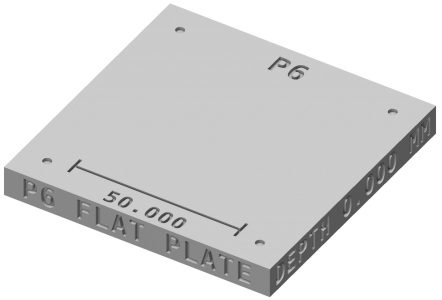


P5 Paraboloid

P5_paraboloid_12x8_d4.step

Opening 24.000 × 16.000 mm oval
Opening 2.4000 × 1.6000 cm
Block 80 × 80 × 9 mm · 8.0 × 8.0 × 0.9 cm

Depth	4.000 mm	0.4000 cm
Area	301.59 mm²	3.0159 cm²
Volume	603.19 mm³	0.60319 cm³
Perimeter	63.46 mm	6.346 cm



P6 Flat plate

P6_flat_plate.step

No cavity
Block 80 × 80 × 9 mm · 8.0 × 8.0 × 0.9 cm

Depth	0.000 mm	0.0000 cm
Area	0.00 mm²	0.0000 cm²
Volume	0.00 mm³	0.00000 cm³

Match the part to its picture, continued



P7 Lobed

P7_lobed_r18_d9.step

Opening R 13.132–22.868 mm, lobed
Opening R 1.3132–2.2868 cm
Block 80 × 80 × 13 mm · 8.0 × 8.0 × 1.3 cm

Depth	9.000 mm	0.9000 cm
Area	1039.45 mm²	10.3945 cm²
Volume	4677.55 mm³	4.67755 cm³
Perimeter	126.82 mm	12.682 cm

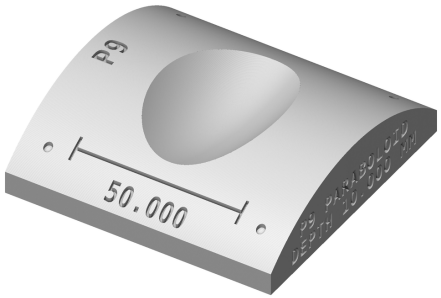


P8 Trough

P8_trough_L40_r9_d8.step

Opening 58.000 × 18.000 mm slot
Opening 5.8000 × 1.8000 cm
Block 80 × 80 × 12 mm · 8.0 × 8.0 × 1.2 cm

Depth	8.000 mm	0.8000 cm
Area	974.47 mm²	9.7447 cm²
Volume	4857.88 mm³	4.85788 cm³
Perimeter	136.55 mm	13.655 cm

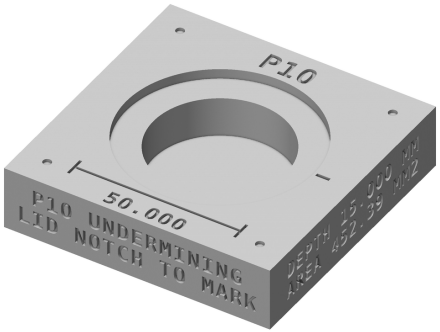


P9 Paraboloid

P9_paraboloid_on_cyl_r60.step

Opening 50.000 × 36.000 mm oval
Opening 5.0000 × 3.6000 cm
On a curved top, R 60 mm · 6.0 cm
Block 80 × 80 × 22 mm · 8.0 × 8.0 × 2.2 cm

Depth	10.000 mm	1.0000 cm
Plan area	1413.72 mm²	14.1372 cm²
Skin area	1446.60 mm²	14.4660 cm²
Volume	7068.58 mm³	7.06858 cm³
Plan perimeter	135.98 mm	13.598 cm



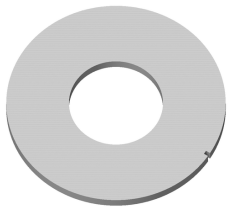
P10 Undermining

P10_undermining_base.step

With a lid seated:
Opening Ø24.000 mm lid hole
Opening Ø2.4000 cm
Block 80 × 80 × 19 mm · 8.0 × 8.0 × 1.9 cm

Depth	15.000 mm	1.5000 cm
Area	452.39 mm²	4.5239 cm²
Volume	16436.81 mm³	16.43681 cm³
Hidden volume	9650.97 mm³	9.65097 cm³
Undermined area	804.25 mm²	8.0425 cm²
Perimeter	75.40 mm	7.540 cm

P10’s lids and undermining: page 4.



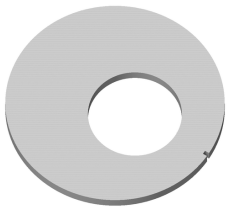
L1 Lid for P10

L1_lid_concentric_r12.step

Hole Ø24.000 mm · Ø2.4000 cm, centred
Disc Ø56 × 3 mm · Ø5.6 × 0.3 cm

Undermining, all round	8.000 mm	0.8000 cm

Notch to P10’s groove. Every clock position: page 4.



L2 Lid for P10

L2_lid_offset6_r12.step

Hole Ø24.000 mm · Ø2.4000 cm, its centre 6.000 mm · 0.6000 cm off-centre, on the notch side
Disc Ø56 × 3 mm · Ø5.6 × 0.3 cm

Least undermining, at 3 o’clock	2.000 mm	0.2000 cm
Most undermining, at 9 o’clock	14.000 mm	1.4000 cm

Notch to P10’s groove. Every clock position: page 4.

Revision B prints have the same shapes and sizes but look different: most of their lettering is missing; the hollow blocks (all but P5 and P6) have round powder holes through their left and right walls, except P9, where the holes come out through its curved top as four open slots in place of the dots; P10 has no index groove beside its pocket; and the lids have no notch.

Every part: file, shape and sizes

Revision C. Each size in mm, with cm beneath it in grey; depth, area and volume in mm as engraved on the parts. Revision B prints read from this table too.

Part	STEP file	How to recognise it	Wound shape (mm)	Opening mm cm	Max depth mm cm	Opening area mm² cm²	Perimeter mm cm	Volume mm³ cm³ = mL
P1	P1_hemisphere_r15.step	19 mm tall; small round bowl, Ø30	Hemisphere, R 15.000	Ø30.000 Ø3.0000	15.000 1.5000	706.86 7.0686	94.25 9.425	7068.58 7.06858
P2	P2_hemisphere_r25.step	29 mm tall, the tallest block; large round bowl, Ø50	Hemisphere, R 25.000	Ø50.000 Ø5.0000	25.000 2.5000	1963.50 19.6350	157.08 15.708	32724.92 32.72492
P3	P3_cone_r20_d10.step	14 mm tall; round opening Ø40, straight sides to a point	Cone, R 20.000, h 10.000	Ø40.000 Ø4.0000	10.000 1.0000	1256.64 12.5664	125.66 12.566	4188.79 4.18879
P4	P4_paraboloid_30x20_d12.step	16 mm tall; large oval 60 × 40	Elliptic paraboloid, a 30.000, b 20.000, h 12.000	60.000 × 40.000 oval 6.0000 × 4.0000	12.000 1.2000	1884.96 18.8496	158.65 15.865	11309.73 11.30973
P5	P5_paraboloid_12x8_d4.step	9 mm tall; small, shallow oval 24 × 16	Elliptic paraboloid, a 12.000, b 8.000, h 4.000	24.000 × 16.000 oval 2.4000 × 1.6000	4.000 0.4000	301.59 3.0159	63.46 6.346	603.19 0.60319
P6	P6_flat_plate.step	9 mm tall; flat top, no cavity	None: flat reference	—	0.000 0.0000	0.00 0.0000	—	0.00 0.00000
P7	P7_lobed_r18_d9.step	13 mm tall; flower-shaped opening, uneven lobes	Lobed paraboloid, R0 18.000, h 9.000; 3-lobe amplitude 0.18 R0, 5-lobe amplitude 0.10 R0 at phase 0.7 rad	R 13.132–22.868, lobed R 1.3132–2.2868	9.000 0.9000	1039.45 10.3945	126.82 12.682	4677.55 4.67755
P8	P8_trough_L40_r9_d8.step	12 mm tall; long slot 58 × 18, rounded ends	Trough: parabolic channel 40.000 long, R 9.000, h 8.000, half-paraboloid ends	58.000 × 18.000 slot 5.8000 × 1.8000	8.000 0.8000	974.47 9.7447	136.55 13.655	4857.88 4.85788
P9	P9_paraboloid_on_cyl_r60.step	22 mm tall; top face curved; oval 50 × 36	Elliptic paraboloid, a 25.000, b 18.000, h 10.000, on a cylinder R 60.000	50.000 × 36.000 oval (plan) 5.0000 × 3.6000	10.000 1.0000	1413.72 14.1372 skin 1446.60 14.4660	135.98 13.598	7068.58 7.06858
P10	P10_undermining_base.step	19 mm tall; stepped round pocket Ø56.4 that the lids drop into	Undermined well: chamber Ø40.000 × 12.000 under a Ø24.000 lid opening	Ø24.000 lid hole Ø2.4000	15.000 1.5000 with a lid	452.39 4.5239	75.40 7.540	16436.81 16.43681
L1	L1_lid_concentric_r12.step	Disc Ø56 × 3, hole centred	Lid for P10: hole Ø24.000, centred	Sets P10's opening and undermining; values on page 4				
L2	L2_lid_offset6_r12.step	Disc Ø56 × 3, hole 6 mm off-centre	Lid for P10: hole Ø24.000, offset 6.000	Sets P10's opening and undermining; values on page 4				

Blocks are 80 × 80 mm; the height is in “How to recognise it”. **Max depth** is from the top surface to the lowest point of the wound (on P9, from the crown of its curved top). **Opening area** and **perimeter** are measured in plan, looking straight down; on P9, whose top is curved, the area measured on the curved surface itself is given as “skin”. **Perimeter** is not engraved on the parts. P10’s depth, area, perimeter and volume are with either lid seated (page 4). Each cm figure is the mm figure with its decimal point moved (1 cm = 10 mm, 1 cm² = 100 mm², 1 cm³ = 1000 mm³ = 1 mL), so it is exactly as precise.

Measure the printed parts with calipers on arrival and use the measured values as truth: every print process shrinks a little, and the measured number is the one that matters.

P10 undermining set

P10 is a Ø40.000 × 12.000 mm (Ø4.0000 × 1.2000 cm) chamber under a Ø56.400 × 3.000 mm (Ø5.6400 × 0.3000 cm) pocket. A lid drops into the pocket, flush with the top, and its Ø24.000 mm (Ø2.4000 cm) hole becomes the visible wound; the chamber beyond the hole's edge is undermined. The two lids give the same depth, area and volume and differ only in where the undermining runs.

	P10 with L1 seated (L1_lid_concentric_r12.step)	**P10 with L2 seated** (L2_lid_offset6_r12.step)
Lid hole	Ø24.000 mm · Ø2.4000 cm, centred	Ø24.000 mm · Ø2.4000 cm, centre 6.000 mm · 0.6000 cm toward 3 o'clock
Max depth	15.000 mm · 1.5000 cm (top skin to chamber floor)	15.000 mm · 1.5000 cm (top skin to chamber floor)
Opening area	452.39 mm² · 4.5239 cm²	452.39 mm² · 4.5239 cm²
Opening perimeter	75.40 mm · 7.540 cm	75.40 mm · 7.540 cm
Total volume	16436.81 mm³ · 16.43681 cm³	16436.81 mm³ · 16.43681 cm³
Visible from above	6785.84 mm³ · 6.78584 cm³	6785.84 mm³ · 6.78584 cm³
Hidden under the lid	9650.97 mm³ · 9.65097 cm³	9650.97 mm³ · 9.65097 cm³
Undermined area	804.25 mm² · 8.0425 cm²	804.25 mm² · 8.0425 cm²
Undermining extent	8.000 mm · 0.8000 cm, all round	2.000 mm · 0.2000 cm at 3 o'clock, to 14.000 mm · 1.4000 cm at 9 o'clock

Without a lid, the chamber alone, measured from the pocket floor 3.000 mm · 0.3000 cm below the top: opening Ø40.000 mm · Ø4.0000 cm, area 1256.64 mm² · 12.5664 cm², depth 12.000 mm · 1.2000 cm, volume 15079.64 mm³ · 15.07964 cm³.

Undermining extent at each clock position

From the edge of the lid's hole outward to the chamber wall, along a line from the hole's centre, as a probe would measure it.

Clock	**12**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
L1 (mm)	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000	8.000
L1 (cm)	0.8000	0.8000	0.8000	0.8000	0.8000	0.8000	0.8000	0.8000	0.8000	0.8000	0.8000	0.8000
L2 (mm)	7.079	4.313	2.578	2.000	2.578	4.313	7.079	10.313	12.970	14.000	12.970	10.313
L2 (cm)	0.7079	0.4313	0.2578	0.2000	0.2578	0.4313	0.7079	1.0313	1.2970	1.4000	1.2970	1.0313

Orientation. 12 o'clock is the edge of P10 nearest the "P10" engraved on its top; 6 o'clock is the side with the 50.000 key line. On revision C parts, seat the lid with the notch in its rim at the index groove on P10's top, which is at 3 o'clock. Revision B parts have no groove, no notch and no legible "P10": take 6 o'clock as the side with the key line, and seat L2 with its hole nearest 3 o'clock. L1 reads the same at every position.

Scale references on every block's top

Key line: 50.000 mm · 5.0000 cm between the centres of its two end ticks. **Fiducial dots:** four Ø2.500 mm · Ø0.2500 cm dots at the corners of a 64.000 mm · 6.4000 cm square, centre to centre. Revision B P9 has no dots: its powder vents cut them away and left four open slots in its top near the corners, so use it for depth, area and volume only, not as a scale reference.

Where the numbers come from

Hemisphere: $V = \frac{2}{3}\pi R^3$, $A = \pi R^2$. Cone: $V = \frac{1}{3}\pi R^2 h$, $A = \pi R^2$. Elliptic paraboloid: $V = \frac{1}{2}\pi abh$, $A = \pi ab$. Lobed paraboloid, $r(\theta) = R0(1 + \Sigma\, a\cos(m\theta + \varphi))$, one term per set of m lobes with amplitude a and phase φ: $V = \frac{1}{4}\pi hR0^2(2 + \Sigma a^2)$, $A = \frac{1}{2}\pi R0^2(2 + \Sigma a^2)$. Trough, channel length L: $V = (4/3)hRL + \frac{1}{2}\pi hR^2$, $A = 2RL + \pi R^2$. P10 with a lid of hole radius r and thickness t: $V = \pi r^2 t + \pi R^2 h$, depth = t + h. Perimeters are integrated numerically from the same outlines.